

POWERFACTORY

PowerFactory 2024

Technical Reference

HVDC Models in PowerFactory

HVDC LCC Monopole CIGRE

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 Introduction

This document describes a High-Voltage Direct Current (HVDC) line commutated converter (LCC) simulation model as developed in *PowerFactory* and delivered as a general template (named *HVDC LCC Monopole CIGRE*). The model is based on published literature [3] and represents a two-terminal monopole HVDC-LCC link. The HVDC system itself is extracted and made available within this template, while excluding all AC network components described within the fore mentioned reference and belonging to the supply networks of both sides of the link. The model can be used for load flow, RMS- and EMT-Simulation analyses.

2 Overview of the *HVDC LCC Monopole CIGRE* Model

The single line diagram of the *HVDC LCC Monopole CIGRE* model is shown in Figure 2.1. The model consists of a detailed dc link representation where each station is realised using two individual 6-pulse converter elements (based on the Rectifier/Inverter element *ElmRec*). A detailed representation of the DC system is available including a lumped T-Line Model with parameterisation intended for the representation of a DC cable system. On the AC side, dedicated converter transformers are modelled, two for each DC converter station. Furthermore, shunt compensation (mechanically switched capacitors - MSC) and AC filters are installed at each station.

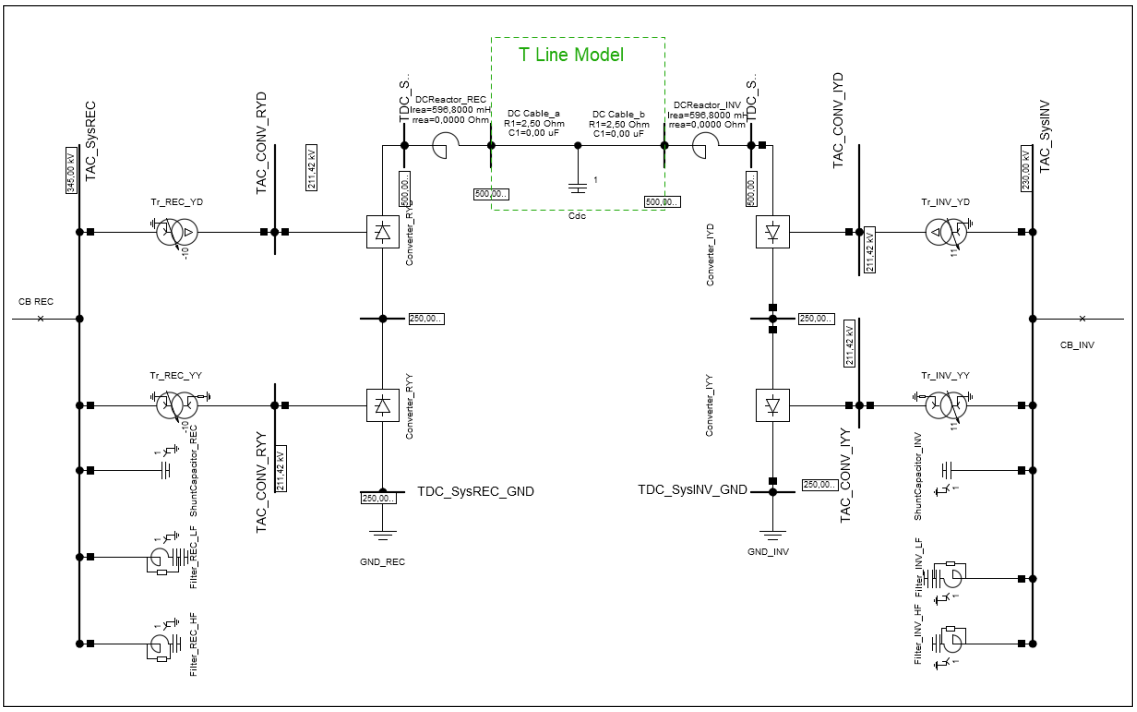


Figure 2.1: HVDC-LCC Single Line Diagram

The ratings of the provided *HVDC LCC Monopole CIGRE* template are given below:

- Nominal power, Converter Station 1 and 2: 1000 MW
- Nominal voltage, Station 1: 345 kV

- Nominal voltage, Station 2: 230 kV
- Shunt Compensation (including AC filters), Station 1: 627 Mvar.
- Shunt Compensation (including AC filters), Station 2: 627 Mvar.

Note: The above configuration can be customised according to the user requirements by following the instructions in this User Manual.

The relevant functions of the *HVDC LCC Monopole CIGRE* model are:

- Load Flow model - regulation of active power flow through the DC link
- Dynamic Model - regulation of direct current flow through the DC link
- Dynamic Model - detailed representation of rectifier/inverter firing angle control
- Dynamic Model - Rectifier controls: constant current and minimum α
- Dynamic Model - Inverter controls: minimum γ and constant current control

The model is designed to support the following calculation fuctions:

- Load Flow calculation,
- Stability (RMS) simulation (see also 3.3),
- Electro-Magnetic Transient (EMT) simulation (see also 3.4).

Refer to published literature [3] in order to understand the types of analyses that can be executed using this model.

For using the model with any other calculation functions, appropriate considerations must be taken in order to obtain plausible behaviour. The model equations and control diagrams are open and available for inspection, hence the user can apply further modifications to them, if so required.

2.1 *HVDC LCC Monopole CIGRE* Model

The HVDC system, excluding the AC network components described within the reference[3], is extracted and made available within this template. It thus enables users to easily deploy the HVDC link within any other power system. The model is not specific to any real-world HVDC installation. Control system parameters are intentionally not specified within [3]. The template contains one set of typically used control parameters, commonly found in various references [1] [4].

The HVDC system corresponds to a 12-pulse monopolar configuration. Dedicated *YY* and YD transformers connect the AC system to the individual 6-pulse converters. The DC system has a voltage rating of 500 kV. The DC link parameters are typical for a DC cable system. The DC cable is represented using a *T model*, where a lumped cable capacitance is placed in the middle of the link.

The commutating bus on the rectifier side is named *TAC_SysREC*. The commutation reactance X_{cr} is defined below, where Xt_{pu} is the per unit leakage impedance of the converter transformer, U_{conv} is the nominal voltage at the converter terminals and St is the rated power of the converter transformer:

$$X_{cr} = Xt_{pu}\frac{U_{conv}^2}{St} = 13.5845Ohm \tag{1}$$

The commutating bus on the inverter side is named *TAC_SysINV*. The commutation reactance X_{ci} is defined below:

$$X_{ci} = 13.3152Ohm \tag{2}$$

The dynamic control system is based on references [1] and [4]. Several modifications and enhancements have been applied for *PowerFactory* specific purposes. The system is designed in *PowerFactory* using a high-level control system architecture based on a Composite Model Frame as shown in Figures 2.2. The figure shows how the converter models are linked with the dynamic controller models, phase-measurement devices and voltage & current measurement devices.

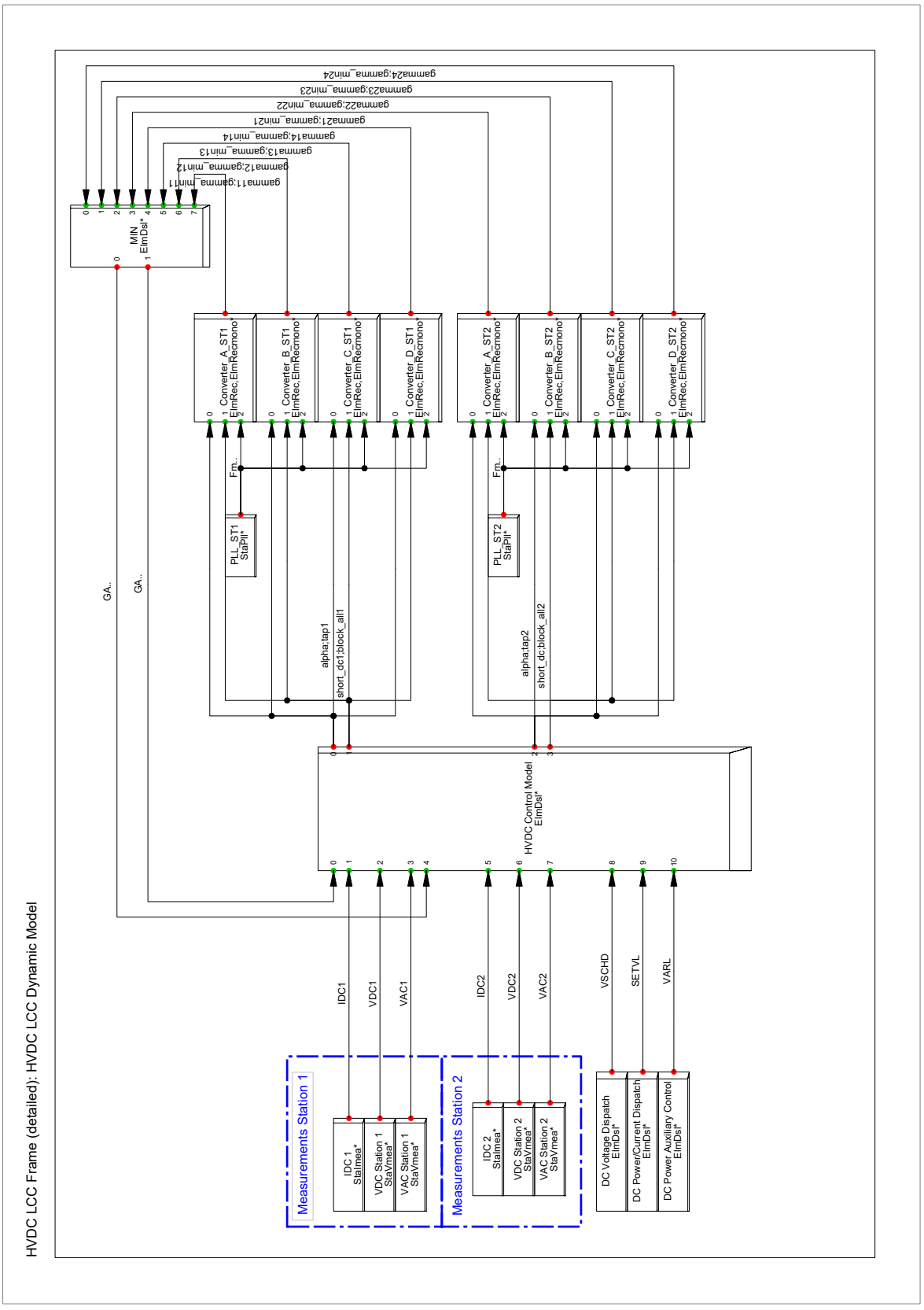


Figure 2.2: High level diagram of HVDC Controls (Composite Model Frame)

The combined steady-state V-I characteristic of the HVDC link (normal power flow direction) is shown in Figure 2.3. This characteristic is identified via analysis of the dynamic controller model structure and functionality. The rectifier characteristic is defined by the natural voltage (NV) characteristic and the constant current control (CC) segments. The inverter characteristic is defined by the extinction angle control (EAC) and the constant current (CC) control segments. The intersection point of the two characteristics represents the steady state operating point. The current setpoints for the rectifier and inverter current controllers are $Iset = CORDER$ and $Iset = CORDER - CMARG$ respectively, with $CMARG$ being a model parameter and $CORDER$ representing an internal control signal computed on the basis of the current order input signal $SETVL$ of the HVDC Controls and, in case of low AC side voltage, a voltage dependent current order limiter (VDCOL) function. The dashed green and blue characteristics are exemplary and may differ depending on the parameterisation of the VDCOL function.

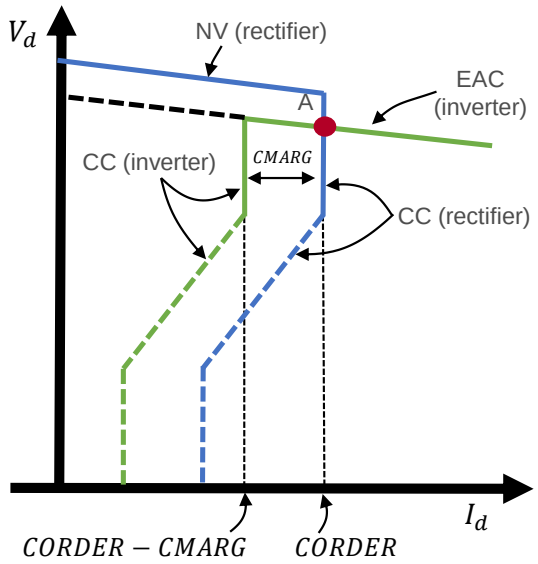


Figure 2.3: Combined steady state V-I characteristic of the HVDC control system

The model includes a pre-configured VDCOL function acting in case of reduced sending-end station DC voltage. The effect of the VDCOL on the combined V-I characteristic, along with the default parameter values are shown in Figure 2.4. In the figure, the VDCOL characteristic is marked by the dashed red curve. The actual current setpoint is selected as the minimum between the steady state current setpoint (i.e. $CORDER$) and the direct current setpoint output of the VDCOL function found for a given DC voltage.

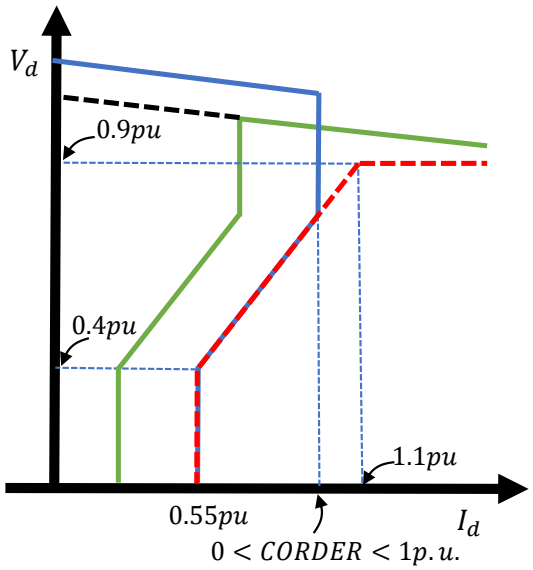


Figure 2.4: Effect of VDCOL function on the combined V-I characteristic

Figure 2.5 shows the actual HVDC controller, as implemented in *PowerFactory* using DIgSILENT Simulation Language (DSL). The core functionality is defined in the dashed-blue rectangle, which contains two sections: “Rectifier Controls” and “Inverter Controls”. The dynamic controller is operating on p.u. quantities, hence changes applied to the rated values of the base model do not require a re-parameterisation of the dynamic model.

The “Rectifier Controls” section is responsible with the direct current regulation at rectifier side. Under normal conditions, this controller regulates the DC current to the reference ”Id_ref”, which is calculated from the load flow solution. In the event of a severe drop in the DC voltage, the current reference is reduced through the VDCOL (voltage-dependent current-order limiter). In case the controller is saturated to its minimum firing angle value (defined by $\alpha_{min} = \pi - BETAR_MAX$, where $BETAR_MAX$ is a model parameter), then the rectifier operates on the NV characteristic.

The “Inverter Controls” section displays the inverter-side controller. Under normal conditions the controller regulates the measured extinction angle $GMES$ to $GMIN$ (where $GMIN$ is a model parameter). In the event of the rectifier not being able to control the direct current, an automatic switching between the EAC and CC modes occurs within the inverter controller , thus leading to the inverter operating in a constant current control (CC) mode, with reduced setpoint current ($Iset = CORDER - CMARG$).

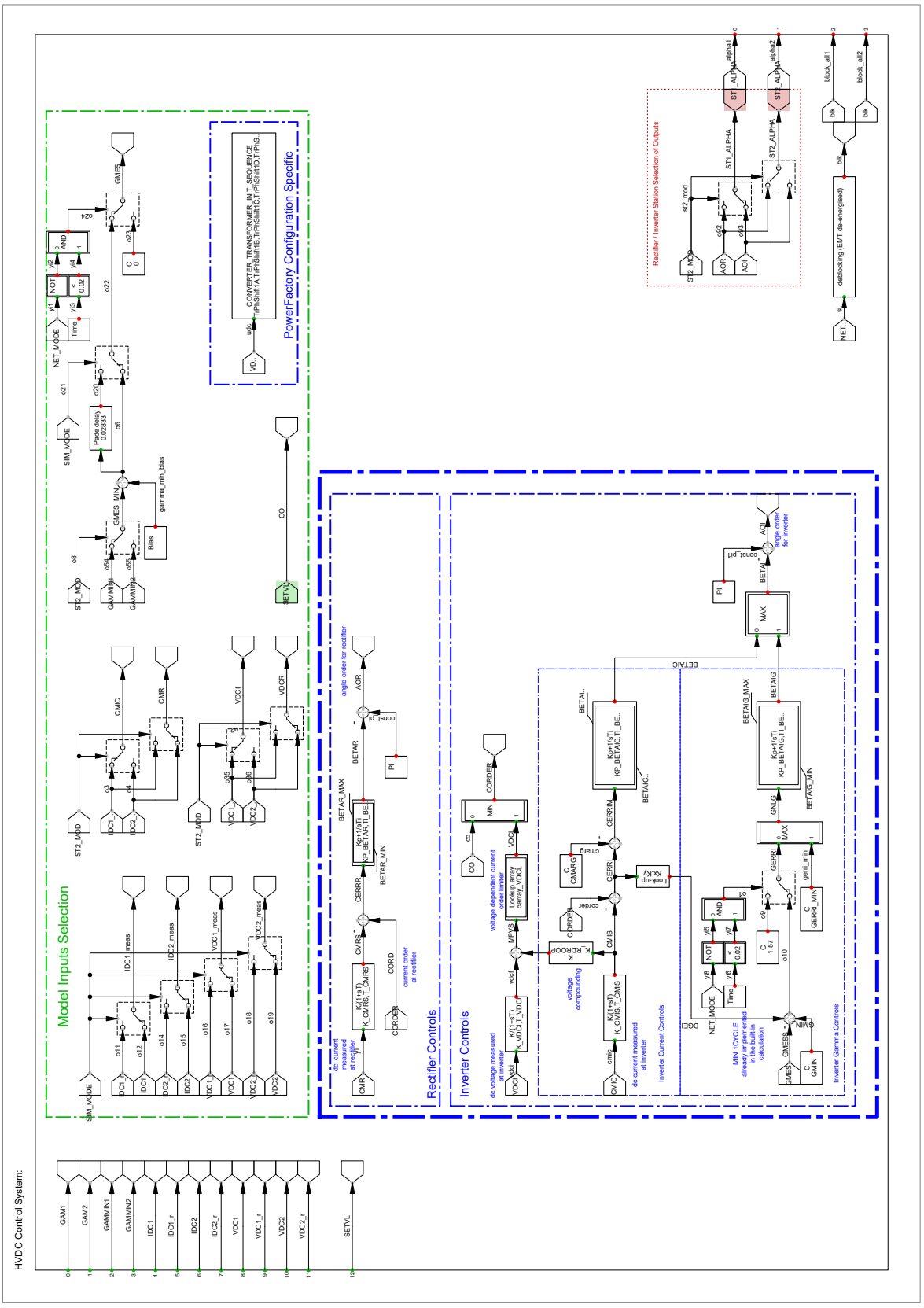


Figure 2.5: Dynamic Controller Block Diagram (DSL Model)

The dynamic HVDC model (including controls) supports both RMS- and EMT-simulation domains. The model is capable of initialising in RMS- and EMT-Simulations based on a pre-existing load flow solution (i.e. with the HVDC link energised during load flow). For EMT-Simulation it automatically supports a simplified energisation procedure (i.e. with the HVDC link initially disconnected during load flow calculation and a subsequent and simultaneous closing of station AC breakers (*CB_REC* and *CB_INV*) at a user-defined time during simulation).

For RMS-Simulations the converter model uses an average value AC system representation[2]. Power balance equations apply between the AC and DC sides, whereas on the DC side a user defined DC circuit is supported.

The combined AC-DC system may sometimes be operationally challenging whenever considerably pronounced AC/DC system interactions exist due to various reasons [3]. For example, one could expect operationally challenging HVDC links whenever one, or more of the following conditions occur:

- a weak rectifier/inverter AC system (low SCR or low inertia),
- resonant points of the AC or DC systems are found close to the AC fundamental frequency or to specific harmonic frequencies.

Whenever such cases occur, a strong coupling between the AC and DC side components can be observed, for which an adequate modelling approach is also required. It is also observed that current paths involving the DC link capacitance and AC side components (e.g. shunt capacitances or filters) may not be accurately represented in RMS type simulations due to the fact that the AC side components are only represented using steady state phasor equations at fundamental frequency. It follows that any direct current dynamics belonging to these current paths will be characterised by inaccurate transient behaviour, even if a highly detailed modelling approach of the DC link (including DC capacitance, DC filters, etc.) would be adopted. In order to avoid unplausible RMS-simulation trajectories, it is safer in these specific cases to remove such current paths completely by applying various simplifications to the DC circuit (e.g. by removing the DC link capacitance). For example, in the case of the AC system configuration proposed in the reference literature [3]), it may be relevant to simplify the lumped T-line model by deactivating the DC-link capacitance for RMS type studies.

For EMT-Simulations the converter model is represented using a detailed 6-pulse thyristor bridge model (that includes the six thyristor valves and their associated snubber circuits)[2]. The firing of each thyristor is internally computed on the basis of the firing angle command (from the controls) and an internally computed synchronising angle. For computation of the latter, a PLL model is used that identifies the commutating bus frequency. The firing instant is computed relative to the internal synchronising angle “phiref”, which varies at the rate of the frequency signal that is connected to the converter model (PLL frequency). For a correct operation of the thyristor firing controller, the model requires a correct value of the commutation reactance to be entered. The EMT-Simulation model is adequate to be used for a larger array of generic system studies, ranging from transient stability analysis, control system interactions and up to sub-synchronous torsional-interaction studies.

Note: The *HVDC LCC Monopole CIGRE* model intentionally does not include any additional control logic that handles abnormal operation of thyristor valves. Especially in the case of prolonged commutation failure events, the dynamic controls do not apply (neither in RMS nor in EMT) any converter blocking or thyristor bridge bypassing actions in order to remove the fault. Whenever such studies/events are in focus, consider including additional control functionality that may adequately represent actual converter converter protection actions.

2.2 Base Model Configuration

This section contains information about the configuration of the HVDC model.

2.2.1 Adjusting Converter Parameters

The HVDC model ratings are available via several *PowerFactory* objects:

- The Rectifier/Inverter elements *ElmRec* and types *TypRec*
- The dynamic control DSL model *HVDC Dynamic Controller*
- The DC circuit components e.g. DC cable model, DC buses
- The AC circuit components e.g. the shunt and filter elements, AC buses

2.2.2 Adjusting the rated AC power

The rated AC power must be adjusted individually for each converter station. For each station, this is done via the two corresponding Rectifier/Inverter types (*TypRec*), under *Basic Data* → *Rated Active Power*. After adjusting the rated power, make sure to adequately adapt the active power dispatch for load-flow calculation in the leading Rectifier/Inverter element (*ElmRec*) of the sending-end station (e.g. Converter_RYD), under *Load Flow* → *Power-Setpoint*.

2.2.3 Adjusting the rated DC voltage

The rated DC voltage must be adjusted individually for each converter station. For each station, this is done via the two corresponding Rectifier/Inverter types (*TypRec*), under *Basic Data* → *Rated DC-Voltage(DC)*. Furthermore, all elements of the DC circuit must be correspondingly updated to reflect this change. After adjusting the rated DC voltage, make sure to adequately adapt the DC voltage dispatch for load-flow calculation in the leading Rectifier/Inverter element (*ElmRec*) of the receiving-end station (e.g. Converter_IYY), under *Load Flow* → *Voltage Set-point*.

2.2.4 Adjusting other converter ratings

Similar to the change of the rated AC power described above, many other converter ratings may be changed within the Rectifier/Inverter type objects (*TypRec*) using the various available tabs.

2.3 Dynamic Model Configuration

The dynamic simulation is initialised in *PowerFactory* on the basis of a steady state solution obtained via a load flow calculation. This means that the starting point for the dynamic simulation will always be the load flow solution. It is therefore important to check that the load flow configuration of all calculation relevant elements is correct.

The composite model which contains the links to all dynamic model parts (DSL model, measurement devices, etc) can be accessed via the Data/Model Manager.

- Via Data or Model Manager: Search the corresponding composite model, right click and select “Edit”.
- After opening the composite model dialog the different components of the HVDC dynamic model can be accessed via a double click in the second column.

Note: The models which are available in *PowerFactory* as templates are configured with a set of default parameters. Whenever necessary, the user is encouraged to adjust all relevant parameters (i.e. power rating, voltage level, reactive power limits and all parameters of the base/dynamic model) in order to represent a specific unit.

The dynamic model can be configured via model parameters. These parameters are displayed after opening the dialog of the DSL model *HVDC Control System* on the “General” page. The DSL *HVDC Control System* contains scalar and array parameters as well (i.e. a list of parameters). The array characteristics can be found on the second tab of the edit dialog, tab named “Advanced 1”.

3 Model Usage Guidelines

3.1 Model deployment within a power system model

The model is delivered as a general template for the purpose of simple network deployment and ease of use. A new model (based on the general template) can be added to, modified in and removed from an active project by following the instructions provided in the *PowerFactory* User Manual, Chapter “Libraries”, section “Templates Library”, subsection “General Templates”.

Upon successful deployment, the model is pre-configured and can be directly used in all supported calculation functions.

3.2 Load Flow Analysis

In load flow calculation, the Rectifier/Inverter elements (*ElmRec*) have the purpose of controlling both the rectifier and the inverter stations such that various load flow objectives are met. For a comprehensive description of the load flow control modes of the *ElmRec* model please refer to the corresponding Technical Reference documentation. The rectifier is by default set up to regulate dc power at the sending-end station. The inverter is set by default to regulate direct voltage at the receiving-end station. This operation mode is regarded as steady state, normal operation mode.

Note: Changing the power flow direction is not supported with this template version.

3.3 Dynamic RMS-Simulation

The model is adequate for RMS-simulation studies (balanced/unbalanced) considering the following simulation settings:

- Maximum time step: 0.01 s (10 ms)

- Minimum time step: 0.001 s (1 ms)

Note: The above recommends that the simulation time step settings should be within the fore-mentioned range. There is no known restriction in terms of the fixed/variable time step option, i.e. both fixed and variable simulation step size adaptation settings are possible. Nevertheless, the user is advised to use the fore mentioned recommended minimum time step, due to the existence of fast reacting HVDC controllers.

Note: The model operates on positive sequence quantities (feedback signals as well as set-points). The model can be used within an unbalanced RMS simulation, nevertheless, the default operation will remain, surely, as a positive sequence control scheme.

3.4 Dynamic EMT-Simulation

The model is adequate for EMT-simulation studies considering the following simulation settings:

- Simulation time step: 0.0001 s (0.1 ms)

Note: When using automatic time step adaptation option, ensure that the maximum time step is selected not greater than 0.0001 s.

3.5 Dynamic Simulation Options

3.5.1 HVDC Re-dispatch: changing direct current setpoint

The direct current setpoint of the sending-end station can be changed during the simulation. This can be done via parameter events, as follows:

- The input signal “SETVL” of model “HVDC Control System” represents the external current order setpoint of the dynamic model. Internally, the control model assigns the control variable $CORDER$ to be equal to $SETVL$.
- Create a parameter event (*EvtParam*) at a certain simulation time and set the target element to be the DSL model “HVDC Control System”. Set the *Name of Variable* to be “SETVL”. Set a wished *Value* in the dialog.
- Run a simulation in order to observe the re-dispatch action.

4 References

[1] FGH controls for the HVDC benchmark model study. Technical report, FGH report presented to CIGRE WG 14.02, 1988.

[2] DIgSILENT GmbH, Heinrich-Hertz-Strasse 9, 72810 Gomaringen / Germany. *Technical Reference Rectifier/Inverter*, 2023.

[3] M. Szechtman (Convener), T. Wess, and C.V. Thio. First benchmark model for HVDC control studies / Premier modèle de référence destiné à des études portant sur le contrôle-commande de réseaux CCHT. In *Electra Magazine no. 135*. CIGRÉ, 1991.

[4] N. Watson and J. Arrillaga. *Power Systems Electromagnetic Transients Simulation (2nd Edition)*. The Institution of Engineering and Technology, 2018.

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